Rapid Revision Summary

Last-minute review · condensed from 16 papers and mark schemes, Jan 2019 – Jun 2026

THE PAPER

2 hours · 100 marks · 10–11 questions · same specification as Paper 1, but more trig proof, rational curves with tangents/normals, second-derivative "show that", set-of-values inequalities and vector proofs.

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mensuration · AP/GP sums, S_∞ · binomial · quotient rule · cosine rule · sin, cos, tan $(A \pm B)$ · change of base. **Derive** double and triple angles from these when asked.

TOP 12 MARK-LOSERS

- "Use algebra / calculus": answer only = **0**.
- Wrong quadratic + calculator roots = M0A0.
- Degrees when radians asked: both A marks lost.
- Decimals mixed into an **exact** answer.
- Extra solutions in range; rounding an out-of-range value in.
- Unrejected impossible values (negative t, $k = -2$).
- S_∞ used with $|r| > 1$.
- Least n without logs, or inequality not reversed.
- Integration not shown; limits not substituted.
- $<$ vs $\leq$ wrong in inequalities.
- Graph roots with no straight line drawn.
- Missing brackets never recovered.

INEQUALITIES AND ROOTS

SET OF VALUES

Critical values → sketch → inside (< 0) or outside (> 0) → overlap for "both".

Real roots $b^2 - 4ac \geq 0$ · real unequal > 0 · none < 0.

Don't forget $k \neq 0$.

- $2kx^2 + 5kx + 5k - 3 = 0$ real $\Rightarrow 3k(8 - 5k) \geq 0 \Rightarrow 0 < k \leq 8/5$.
- Real x for all y: quadratic in x, discriminant ≥ 0.

A, B

$\alpha + \beta = -b/a \cdot \alpha\beta = c/a \cdot \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) \cdot (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

New equation: x^2 − (sum)x + product = 0, integer coefficients

$2x^2 + (6 + 2p)x + 2p = 0$: $(\alpha - \beta)^2 = 9 + 2p + p^2$; $\alpha - \beta = 3 \Rightarrow p = 0$ or -2.

LOGS AND SURDS

- One base first: $\log_b a = 1/\log_a b$; $\log_{a^0} x = (1/n) \log_a x$.
- $\log_4 x + \log_{16} x + \log_2 x = 10.5 \Rightarrow (7/4) \log_2 x = 10.5 \Rightarrow x = 64$.
- $3 \log_3 x - 8 \log_x 3 = 10$: $u = \log_3 x \Rightarrow (3u + 2)(u - 4) = 0 \Rightarrow x = 81, 3^{-2/3}$.
- Surds: conjugate; show every line; aim for the stated form.

SERIES

FORMULAE

AP: $a + (n - 1)d$ · $(n/2)[2a + (n - 1)d]$

GP: ar^{n-1} · $a(1 - r^n)/(1 - r)$ · $S_\infty = a/(1 - r)$, $|r| < 1$

GP terms: $(\text{middle})^2 = \text{first} \times \text{third}$

- $ar^2 = 10$, $ar^6 = 40 \Rightarrow r = -\sqrt{2}$ (second term negative), $a = 5$; $|r| > 1 \Rightarrow$ no S_∞.
- $2r^2 + 5r + 2 = 0 \Rightarrow r = -½ \Rightarrow a = 96$, $S_\infty = 64$.
- Least n: inequality in r^n → logs → flip for $\log r < 0$ → round up.

BINOMIAL

- $(a + bx)^n = a^n(1 + bx/a)^n$, $|x| < |a/b|$.
- $3/(9 - 3x) = ⅓ + x/9 + x^2/27 + x^3/81$, $|x| < 3$.
- Multiply by another factor, then integrate term by term to the stated s.f.

GRAPHS

- Rearrange to "plotted curve = line"; draw the line; read x to 1 d.p. (2.0 not 2).
- Region R: shade, test vertices; integer points if asked.

RATIONAL CURVES

RATIONAL CURVE

For $y = (ax + b)/(cx + d)$:

Asymptotes $\mathbf{x = -d/c}$, $\mathbf{y = a/c}$ · intercepts $(-b/a, 0)$, $(0, b/d)$

$dy/dx = (ad - bc)/(cx + d)^2$ · tangent ∥ gradient m: $(ad - bc)/(cx + d)^2 = m$

Normal: $-1/(dy/dx)$. Sketch: both branches, no crossing.

- $(2x - 1)/(x + 4)$: $y = 2$, $x = -4$; $(½, 0)$, $(0, -¼)$; $y' = 9/(x + 4)^2$; ∥ $y = x \Rightarrow P(-1, -1)$, $Q(-7, 5)$.
- No stationary points: $2e^{2x}(x^2 + x + 1)$, discriminant $-3 < 0$.

TRIGONOMETRY

DERIVE AND LEARN

$\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A \cdot \sin 2A = 2 \sin A \cos A \cdot \tan 2A = 2t/(1 - t^2)$

$\cos 3A = 4\cos^3 A - 3\cos A \cdot \cos^4\theta - \sin^4\theta = \cos 2\theta$

$\cos P + \cos Q = 2 \cos((P + Q)/2) \cos((P - Q)/2) \cdot \cos^3\theta = (\cos 3\theta + 3\cos \theta)/4$

- $3 \tan A = 2 \sin A \Rightarrow \sin A(3 - 2\cos A) = 0 \Rightarrow 0°, 180°, 360°$. **Factorise, never divide.**
- $7\sin^2\theta + \sin\theta \cos\theta = 6 \Rightarrow \tan^2\theta + \tan\theta - 6 = 0 \Rightarrow 63°, 108°, 243°, 288°$.
- $9\cos A - 12\cos^3 A - 2 = 0 \Rightarrow \cos 3A = -2/3 \Rightarrow A = -0.8, -1.3, -2.9 \ (-\pi \le A \le 0)$.
- $\tan(2\theta - 20)° = \sqrt{3} \Rightarrow \theta = 40°, 130°$.
- Exact: $\cos 15° = (\sqrt{6} + \sqrt{2})/4$ from $45° - 30°$.

RADIANS, TRIANGLES, 3D

- Arc $r\theta$ · sector $\frac{1}{2}r^2\theta$ · segment $\frac{1}{2}r^2(\theta - \sin\theta)$.
- $r = 14$, perimeter $49 \Rightarrow$ arc $21 \Rightarrow$ area 147. Two radius-6 circles: $24\pi - 18\sqrt{3}$.
- Line–plane: use projection. Plane–plane: perpendiculars to the common edge.

VECTORS

- Parallelogram: $\overrightarrow{AB} = \overrightarrow{DC}$. Trapezium: one pair parallel, other not.
- Collinear A, Q, R: $\overrightarrow{OR} = \overrightarrow{OA} + \lambda\overrightarrow{AQ}$; zero coefficient $\Rightarrow \lambda$. Example: OR : RB = 3 : 2.
- $|\mathbf{a}| = |\mathbf{b}| = 1$, $\mathbf{a} \perp \mathbf{b} \Rightarrow |p\mathbf{a} + q\mathbf{b}| = \sqrt{(p^2 + q^2)}$.
- Area ratio = base ratio for a common height.

CALCULUS

SECOND-DERIVATIVE "SHOW THAT"

$y = e^{2x}\cos 2x$: $y' = 2y - 2e^{2x}\sin 2x \Rightarrow y'' - 4y' + 8y = 0$.

$y = e^{3x}\sin 2x \Rightarrow y'' - 6y' + 13y = 0$.

- Rates: $V = 6h^3$, $dV/dt = -36$, $V = 384 \Rightarrow h = 4 \Rightarrow dh/dt = -1/8$.
- Sphere: $dV/dt = 40$, $r = 4 \Rightarrow dS/dt = 20$.
- Optimise: $V = \pi r(350 - r^2) \Rightarrow r = 10.8$, $h = 21.6$; $V'' < 0 \Rightarrow$ max.
- Kinematics: $s = 3 + 8t + t^2 - t^3/3 \Rightarrow v = 0$ at $t = 4 \Rightarrow s = 89/3$. Reject $t < 0$.
- Small change: $3\% \Rightarrow \delta x = 0.03x$.

INTEGRATION

AREAS AND VOLUMES

Area $\int y\,dx$ · between $\int(\text{upper} - \text{lower})\,dx$

Volume $\pi\int y^2\,dx$ (x-axis) · $\pi\int x^2\,dy$ (y-axis)

- Intersections first, algebraically.
- Square trig with $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$.
- $\pi\int e^{6x}\,dx = (\pi/6)e^{6x}$; keep π, e, ln exact.

FINAL 60 SECONDS

- Working shown? Exact or 3 s.f.? Degrees or radians?
- Every root checked, every extra rejected with a reason.
- Graph: line drawn. Integral: expression + limits visible.